

Daniel Herzberg

- > Principal Mechanical Engineering Manager
- > Certified SolidWorks Expert
- > SolidWorks Champion and Educator
- > AI & CAD Automation Specialist

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PROFESSIONAL SUMMARY

Principal Mechanical Engineer and Engineering Manager with a proven record of leading high-performing multidisciplinary teams, integrating AI-driven automation into the design process, and developing robust mechanical systems for robotics, industrial automation, and medical devices. Known for streamlining product development workflows using custom automation tools to accelerate design throughput and eliminate rework. Accomplished SolidWorks Expert and community leader, driving innovation, quality, and speed across complex product lifecycles.

EXPERIENCE

January 2020 – Present

Goddard, Inc. Beverly, MA

Principal Mechanical Engineer & Engineering Manager January 2024 – Present

- Leading and mentoring a multidisciplinary mechanical engineering team of 5, driving technical excellence and employee growth while maintaining high delivery standards.
- Integrating AI and automation tools within SolidWorks and PLM systems, cutting repetitive documentation tasks by 50% and accelerating design reviews by up to 40%.
- Championing CAD process standardization and automating error-checking, reducing rework and improving design release accuracy.
- Collaborating with electrical, software, and project management teams to deliver the next generation of automated package-handling robotic workcells to our industrial automation clients.

Senior Mechanical Engineer January 2020 – December 2023

- Designed and delivered high-reliability systems for industrial automation, robotics, and medical applications, improving ergonomics and long-term durability.
- Designed solutions that increased efficiency and reliability of equipment used by 300+ client fulfillment centers worldwide, while also improving ergonomics and performance.
- Conducted verification and validation testing on medical device which aims to improve quality-of-life of wheelchair-bound users.
- Led a team of ~10 to integrate acquired machine designs into a client's PLM system, saving hundreds of engineering hours through automation and workflow optimization. Developed and implemented SolidWorks automation utilities, reducing manual cleanup and documentation time by up to 50%.
- Managed CAD and PDM administration, training, and best-practice implementation to elevate model integrity and collaboration.

April 2015 – November 2019

Jenike & Johanson, Inc. Tyngsboro, MA

Project Manager

- Managed full-cycle design projects addressing bulk solids flow challenges for global clients, saving customers millions annually through improved uptime and safety.
- Conducted material flow testing and analysis to design reliable bulk solids handling systems tailored to each client's operational requirements.
- Visited client facilities worldwide to diagnose flow issues, collect samples, and collaborate with operators on practical, data-driven solutions.
- Modeled and simulated systems using DEM, FEA, CFD, and wear analysis; produced detailed industrial-scale SolidWorks drawings and documentation through all project phases.

December 2012 – March 2015

Philips Healthcare Andover, MA

Mechanical Design Engineer, Ultrasound Systems

- Led mechanical development of multiple electro-mechanical subsystems and fixtures for premium ultrasound devices from concept to release.
- Defined design specifications, and environmental testing protocols, managed prototype validation, and ensured regulatory compliance under FDA design controls.

- Designed mechanical components and analyzed for manufacturability via many different methods, including injection molding, metal casting, machining, PCB construction and additive manufacturing.

PRIOR EXPERIENCE

August 2011 – March 2015

Owner & Lead Design Engineer at MegaHertz Modeling & Design Cambridge, MA

October 2010 – April 2012

Systems Engineer at Naval Surface Warfare Center Dahlgren, VA

June 2008 – August 2010

Technical Support Engineer at SOLIDWORKS Corporation Concord, MA

LEADERSHIP AND COMMUNITY

September 2012 – Present

Boston SolidWorks User Group Boston, MA

Founder and President

- > Organized guest speakers, vendor demos, sponsorships, and meeting locations while expanding group membership.
- > Planned and executed successful annual global user group meetings, coordinating with dozens of user groups, and bringing together SolidWorks experts and vendors from around the world for networking and education.
- > Grew membership by 300% and hosted events with 100+ attendees, advancing CAD education in the New England region.

October 2014 – May 2017

The Possible Project Cambridge, MA

Design & Makerspace Volunteer

- > Developed and facilitated SolidWorks training curriculum for after-school program aimed at providing engineering and entrepreneurial skills to at-risk students.
- > Empowered young people with the skills and mindset needed to succeed in the workforce and become future leaders.

EDUCATION

Worcester Polytechnic Institute Worcester, MA

Bachelor of Science, Aerospace Engineering High Distinction [GPA 3.45]

Minor Degree: Physics

CERTIFICATIONS

QSG Effective Communication Apr 2023

Lean Manufacturing Oct 2014

Certified SolidWorks Expert Jun 2010

Certified SolidWorks Professional Aug 2009

TECHNICAL EXPERTISE

SolidWorks | SolidWorks PDM | AI Automation | PLM
(Enovia, Oracle Agile, etc.) | GD&T | FEA | CFD | DEM |
DFM/DFA/DFX | Robotics | Product Development
Lifecycle | Technical Communication (SolidWorks
Composer) | Rendering (SolidWorks PhotoView360,
Visualize, Keyshot)

SKILLS

- > Automation & AI Integration
- > 3D modelling & 2D drawings
- > Process Optimization, LEAN Methods, Continuous Improvement
- > Leadership
- > Additive Manufacturing – FDM, SLA, post-processing
- > FEA / CFD / DEM
- > Creativity and Analytical Problem-Solving
- > Collaboration
- > Time Management
- > Self-Motivation
- > Injection Molding / Machining / Metal Casting